

行星大气概论作业2

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1. 从 <http://exoplanetarchive.ipac.caltech.edu> 中, 找出一颗位于宜居带(HZ)的行星系统, 并计算出该系统宜居带的内外边界, 并计算行星的平衡温度(可含待定参数)

解: 从网站中找到 K2-3 d 行星及其宿主恒星 K2-3 的性质如下: (Diamond-Lowe et al. 2022)

$$K2-3: T_{\text{eff}} = 3844_{-63}^{+61} \text{ K} \quad L_s = 0.0587_{-0.0019}^{+0.0018} L_0 \quad R_s = 0.546_{-0.016}^{+0.018} R_0$$

$$K2-3d: a_p = 0.2014_{-0.0033}^{+0.0034} \text{ au} \quad e = 0.091_{-0.064}^{+0.120} \quad T_{\text{eq}} = 305.2_{-3.2}^{+3.1} \text{ K}$$

从 Kopparapu et al. (2013) 中有: $S_{\text{eff}} = S_{\text{eff}0} + aT_* + bT_*^2 + cT_*^3 + dT_*^4$ ($T_* = T_{\text{eff}} - 5780 \text{ K}$)

$$d_{\text{HZ}} = \left(\frac{L/L_0}{S_{\text{eff}}} \right)^{0.5} \text{ AU}$$

	$S_{\text{eff}0}$	a	b	c	d
Runaway Greenhouse	1.0512	1.3242×10^{-4}	1.5418×10^{-8}	-7.9895×10^{-12}	-1.8328×10^{-15}
Moist Greenhouse	1.0140	8.1774×10^{-5}	1.7063×10^{-9}	-4.3241×10^{-12}	-6.6462×10^{-16}
Maximum Greenhouse	0.3438	5.8942×10^{-5}	1.6558×10^{-9}	-3.0045×10^{-12}	-5.2983×10^{-16}

代入 T_{eff} 和 L_s 可得宜居带内 (Runaway or Moist) 外 (Maximum) 边界:

$$\text{Runaway: } S_{\text{eff}} = 0.8848, \quad d = 0.06634 \text{ AU}$$

$$\text{可见该行星轨道的近日距 } a - C_p = \frac{a(1-e)}{1+e} = 0.183 \text{ AU}$$

$$\text{Moist: } S_{\text{eff}} = 0.8841, \quad d = 0.06639 \text{ AU}$$

$$\text{和远心距 } a + C_p = \frac{a(1+e)}{1-e} = 0.220 \text{ AU}$$

$$\text{Maximum: } S_{\text{eff}} = 0.2503, \quad d = 0.23456 \text{ AU}$$

确在宜居带内外边界之内

$$\text{该行星的平衡温度 } T_{\text{eq}} = \left(\frac{R_s}{2a_p} \right)^{\frac{1}{2}} \left(\frac{1-A_B}{\epsilon} \right)^{\frac{1}{4}} T_{\text{eff}} = \left(\frac{1-A_B}{\epsilon} \right)^{\frac{1}{4}} \cdot \left(\frac{0.546 \times 6.957 \times 10^8 \text{ m}}{2 \times 0.2014 \times 1.496 \times 10^{11} \text{ m}} \right)^{\frac{1}{2}} \times 3844$$

$$= \left(\frac{1-A_B}{\epsilon} \right)^{\frac{1}{4}} \cdot 305.197 \text{ K}$$

可见网站中给出的 T_{eq} 忽略了反照 A_B 和 ϵ 因子。

2. 根据 NASA EXOplanet Archive 中的参数, 计算 kepler-34 b 这颗行星 (Geometric albedo: 0.4; Bond albedo: 0.3) 的最低和最高平衡温度。利用程序计算行星接受的辐射强度随时间的变化并计算 10 年内, 行星平衡温度的变化。

解: 从 NASA EXO planet archive 网站及 William F. Welsh et al. (2012) Nature Letters 原文中的 Table 1 可获得 Kepler-34 A, Kepler-34 B (stellar binary) 及 Kepler-34 b (planetary) 的相关性质如下:

Kepler-34 A: $T_{A,eff} = 5913 \pm 130 \text{ K}$ $R_A = 1.1618_{-0.0031}^{+0.0027} R_{\odot}$ $M_A = 1.0479_{-0.0030}^{+0.0033} M_{\odot}$

Kepler-34 B: $T_{B,eff} = 5915_{-117}^{+105} \text{ K}$ (from 41-417 DR25 KOI table), $R_B = 1.0921_{-0.0027}^{+0.0032} R_{\odot}$

binary orbit: $a = 0.22882_{-0.00018}^{+0.00019} \text{ AU}$, $e = 0.52087_{-0.00055}^{+0.00052}$ $M_B = 1.0208_{-0.0022}^{+0.0022} M_{\odot}$

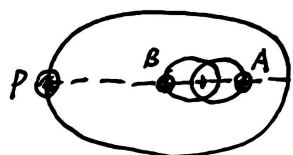
$i = 89.8584_{-0.0073}^{+0.0075} \text{ deg}$, $P = 27.7958103_{-0.000015}^{+0.000016} \text{ days}$

Kepler-34 b: $M_p = 0.220_{-0.010}^{+0.011} M_{Jup}$, $R_p = 0.764_{-0.014}^{+0.012} R_{Jup}$

$a_p = 1.0896_{-0.0009}^{+0.0009} \text{ AU}$, $e = 0.182_{-0.020}^{+0.016}$, $i = 90.355_{-0.018}^{+0.026} \text{ deg}$, $P = 288.822_{-0.021}^{+0.013} \text{ days}$

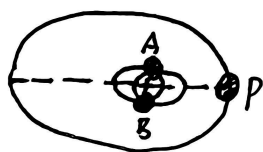
(1) 考虑出现最低及最高平衡温度的极限情形:

Welsh 文中还给出 A, B 两星的 flux ratio $F_B/F_A = 0.8475$, 故最低 T_{eq} 出现在 B 遮挡 A 且行星 P 出现在远心点时:



$$T_{eq,min} = \left(\frac{1-A_B}{\epsilon} \right)^{\frac{1}{4}} \left(\frac{R_B}{2[a_p(1+e) - a_B]} \right)^{\frac{1}{2}} T_{B,eff}$$

其中 a_B 代表 B 到质心的距离, 不妨取 $a_B = a/2 = 0.11 \text{ AU}$. 代入 $A_B = 0.3$, ϵ 取 1 可得 $T_{eq,min} \approx 251.277 \text{ K}$
最高 T_{eq} 出现在 A, B 无遮挡且 P 位于近心点时:



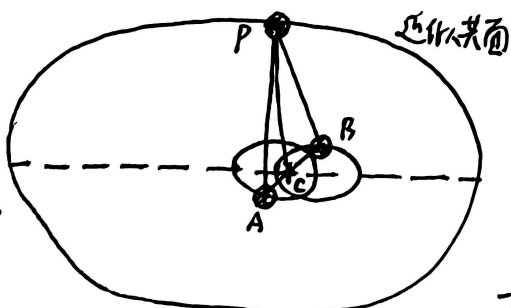
如图, 为简化起见, 忽略双星轨道运动对行星半径的影响,

$$T_{eq,max} = \left(\frac{1-A_B}{\epsilon} \right)^{\frac{1}{4}} \left(\frac{1}{2a_p(1-e)} \right)^{\frac{1}{2}} (R_A^2 T_{A,eff}^4 + R_B^2 T_{B,eff}^4)^{\frac{1}{4}} \approx 348.932 \text{ K}$$

(2) 行星接收的辐射强度随时间的变化及 T_{eq} 随时间的变化. (取 $\epsilon \sim 1$)

主要思路:

"rebound"
可模拟大右
图所示三体
运动



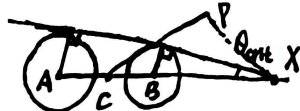
行星接受的总光度 $L = 4\pi R_p^2 \sigma_B T_{eq}^4$

$$= \pi R_p^2 (1-A_B) \left\{ \frac{R_A^2 \sigma_B T_{A,eff}^4}{r_{Ap}^2} + \frac{R_B^2 \sigma_B T_{B,eff}^4}{r_{Bp}^2} \right\}$$

$$\frac{BX}{BX+AB} = \frac{R_B}{R_A} \Rightarrow BX = \frac{AB}{\frac{R_A}{R_B} - 1} \Rightarrow \theta_{crit} = \arcsin\left(\frac{R_B}{BX}\right)$$

CPX 为角: $\theta = \angle CXP = \arccos\left(\frac{CX^2 + PX^2 - CP^2}{2 \cdot CX \cdot PX}\right)$, 当 $\theta < \theta_{crit}$ 时为遮挡

判断遮挡



```

1 import rebound
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from matplotlib.ticker import AutoMinorLocator
5 ✓ [1] 650毫秒

1 sim = rebound.Simulation()
2 sim.units = ('yr', 'AU', 'Msun')
3 sigma = 5.67037442e-08 * (365.25*86400) * (1.49597871e+11)**2 # J/(yr*AU**2*K**4)
4 m_A = 1.0479
5 R_A = 1.1618*695700000/1.49597871e+11
6 m_B = 1.0208
7 R_B = 1.0927*695700000/1.49597871e+11
8 m_P = 0.220*1.8981246e+27/1.98840987e+30
9 R_P = 0.764*71492000/1.49597871e+11
10 a_P = 1.0896
11 e_P = 0.182
12 i_P = 90.355
13
14 M = m_A + m_B
15 a_sep = 0.22882
16 a_A = (m_B / M) * a_sep
17 a_B = (m_A / M) * a_sep
18 omega = np.sqrt(sim.G*M/a_sep**3)
19
20 v_A = omega*a_A
21 v_B = omega*a_B
22 sim.add(m=m_A, x=-a_A, y=0, z=0, vx=0, vy=-v_A, vz=0)
23 sim.add(m=m_B, x=a_B, y=0, z=0, vx=0, vy=v_B, vz=0)
24
25 v_P = np.sqrt(sim.G*(M+m_P)/(a_P*(1-e_P**2)))*(1+e_P)
26 sim.add(m=m_P, x=a_P*(1-e_P), y=0, z=0, vx=0, vy=v_P, vz=0)
27
28 op = rebound.OrbitPlot(sim, figsize=(6,6), unitlabel="[AU]", color=True, periastron=True, xlim=[-1.5,1.0], ylim=[-1.25,1.25])
29 op.fig.savefig("orbit.png", format='png', dpi=300, bbox_inches='tight', pad_inches=0.2)

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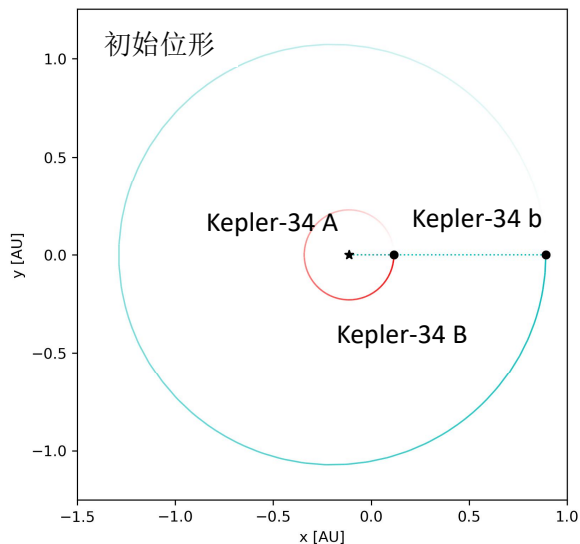
1 sim.integrator = "ias15"
2 dt = 0.00001
3 t_end = 10.0
4 n_steps = int(t_end/dt)
5
6 radiation_intensity = np.zeros(n_steps)
7 equilibrium_temperature = np.zeros(n_steps)
8
9 for i in range(n_steps):
10     sim.integrate(sim.t + dt)
11
12     r_A = np.array(sim.particles[0].xyz)
13     r_B = np.array(sim.particles[1].xyz)
14     r_P = np.array(sim.particles[2].xyz)
15
16     r_AB = np.linalg.norm(r_B - r_A)
17     r_AP = np.linalg.norm(r_P - r_A)
18     r_BP = np.linalg.norm(r_P - r_B)
19
20     r_BX = r_AB/(R_A/R_B-1)
21     r_X = r_B*(1+r_BX/a_B)
22     r_PX = np.linalg.norm(r_X - r_P)
23
24     theta_crit = np.arcsin(R_B/r_BX)
25     theta = np.arccos((r_BX**2+r_PX**2-r_BP**2)/(2*r_BX*r_PX))
26
27     T_A = 5913
28     T_B = 5915
29     L_AP = sigma*np.pi*r_P**2*(1-0.3)*(R_A**2*T_A**4/r_AP**2)
30     L_BP = sigma*np.pi*r_P**2*(1-0.3)*(R_B**2*T_B**4/r_BP**2)
31     if theta<=theta_crit:
32         if r_PX<r_BX:
33             L = L_BP
34         else:
35             L = L_AP

```

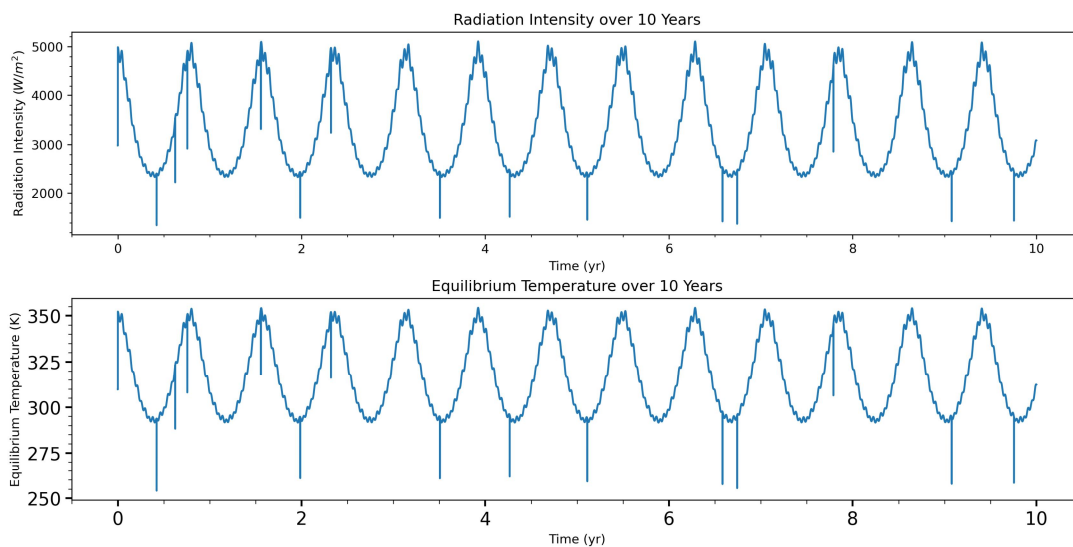
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36     else:
37         L = L_AP + L_BP
38
39         T_eq = np.power(L/(4*np.pi*R_P**2*sigma),0.25)
40         I = L/(np.pi*R_P**2*(1-0.3)) # J/(yr*AU**2)
41         radiation_intensity[i] = I / (365.25*86400 * (1.49597871e+11)**2) # W/m^2
42         equilibrium_temperature[i] = T_eq # K
43
44     time = np.linspace(0, t_end, n_steps)
45     fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12,6), layout='constrained')
46     ax1.plot(time, radiation_intensity)
47     ax1.set_xlabel('Time (yr)')
48     ax1.set_ylabel(r'Radiation Intensity ($W/m^2$)')
49     ax1.set_title('Radiation Intensity over 10 Years')
50     ax2.plot(time, equilibrium_temperature)
51     ax2.set_xlabel('Time (yr)')
52     ax2.set_ylabel('Equilibrium Temperature (K)')
53     ax2.set_title('Equilibrium Temperature over 10 Years')
54     ax1.xaxis.set_minor_locator(AutoMinorLocator())
55     ax1.yaxis.set_minor_locator(AutoMinorLocator())
56     ax2.xaxis.set_minor_locator(AutoMinorLocator())
57     ax2.yaxis.set_minor_locator(AutoMinorLocator())
58     plt.tick_params(which='major',length=5,width=1.5,labelsize='x-large')
59     plt.tick_params(which='minor',length=3)
60     plt.show()
61     fig.savefig('variation.png', format='png', dpi=300, bbox_inches='tight', pad_inches=0.2)

```

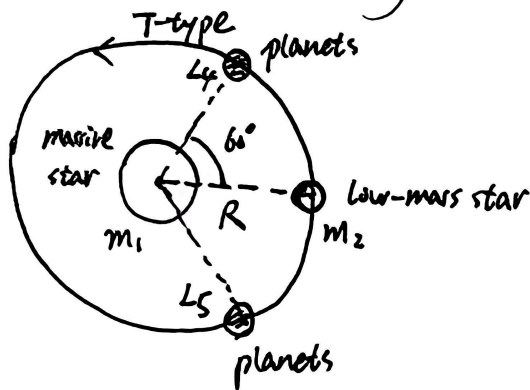


如下图所示，辐射强度和平衡温度主要受双星轨道运动周期以及行星绕系统质心轨道运动的周期共同调制。由于我们对遮挡问题的考虑过于简化，因此辐射强度和平衡温度变化图像中出现很陡地突然下降，即由双恒星共同贡献辐射转为某个恒星单独提供辐射，实际上这个过程应该更为平滑，从外切到内切应有较连续的变化。不过仅从目前的简单模型，我们推断出的平衡温度变化已经较好的符合我们在第一问得到的 $251K \sim 348K$ 的结果。此外，辐射强度约为太阳常数 $1367W/m^2$ 的两倍，这也比较合理，因为该行星的半长轴也在 $1AU$ 附近，但是有两颗温度与太阳接近的恒星共同影响。



3. Discuss: Habitability of T-type planets in binary? Find a criterion

解: T-type planets in binary:



由于 T-type star 位于双恒星平面的三角
平动点 (L_4 或 L_5) 上:

$$L_4 = \left(\frac{1}{2} - \mu, \frac{\sqrt{3}}{2}, 0 \right) R \quad \text{其中 } \mu = \frac{m_2}{m_1 + m_2}$$

$$L_5 = \left(\frac{1}{2} - \mu, -\frac{\sqrt{3}}{2}, 0 \right) R$$

故双恒星-行星之间的构型保持不变, 故
行星可长期接受稳定的辐照.

若通过质光比由 m_1 与 m_2 推断得到恒星光度 L_1, L_2
($L \propto M^{3.5}$)

由于 L_4/L_5 与 m_1, m_2 构成等边三角形, 从而有

$$4\pi R_p^2 \sigma_B T_{eq}^4 = \pi R_p^2 (1 - A_B) \cdot \left(\frac{L_1}{4\pi R^2} + \frac{L_2}{4\pi R^2} \right) \quad (\text{取 } \varepsilon \sim 1)$$

$$\Rightarrow T_{eq} = \left(\frac{1 - A_B}{4\sigma_B} \cdot \frac{L_1 + L_2}{4\pi R^2} \right)^{\frac{1}{4}} = \frac{1}{2} \left(\frac{(1 - A_B)(L_1 + L_2)}{\pi \sigma_B R^2} \right)^{\frac{1}{4}}$$

则对于已知光度 (或质量) 的双恒星, 其 T-type 行星的宜居带范围可通过

$T_{eq, \min} \leq T_{eq} \leq T_{eq, \max}$ 推断出相应的 R 的范围为:

$$\left(\frac{(1 - A_B)(L_1 + L_2)}{16\pi \sigma_B T_{eq, \max}^4} \right)^{\frac{1}{2}} \leq R \leq \left(\frac{(1 - A_B)(L_1 + L_2)}{16\pi \sigma_B T_{eq, \min}^4} \right)^{\frac{1}{2}}$$